实验七.基于 AI 车道巡线

实验过程

1）创建车道巡线的功能包（lanedetection）

cd /userdata/dev_ws/src/originbot

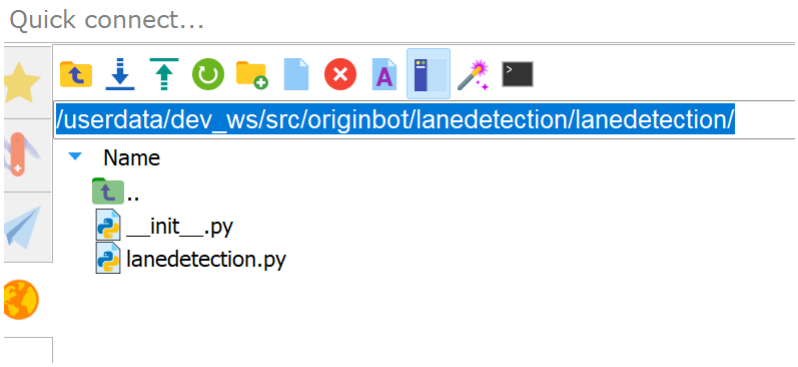
ros2 pkg create --build-type ament_python lanedetection

lanedetection 就是新建功能包的名称

```
root@ubuntu:/userdata/dev_ws# cd /userdata/dev_ws/src/originbot
root@ubuntu:/userdata/dev_ws/src/originbot# ros2 pkg create --build-type ament_python lanedetection
going to create a new package
package name: lanedetection
destination directory: /userdata/dev_ws/src/originbot
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['root <root@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_python
dependencies: []
creating folder ./lanedetection
creating ./lanedetection/package.xml
creating source folder
creating folder ./lanedetection/lanedetection
creating ./lanedetection/setup.py
creating ./lanedetection/setup.cfg
creating folder ./lanedetection/resource
creating ./lanedetection/resource/lanedetection
creating ./lanedetection/lanedetection/__init__.py
creating folder ./lanedetection/test
creating ./lanedetection/test/test_copyright.py
creating ./lanedetection/test/test_flake8.py
creating ./lanedetection/test/test_pep257.py
```

2）创建车道巡线的节点

将车道巡线的源程序 lanedetection.py 上传到 dev_ws/src/originbot/lanedetection/lanedetection/下



修改 dev_ws/src/originbot/lanedetection/setup.py

增加’lanedetection =lanedetection.lanedetection:main’,

注：包括标点符号在内，全部是英文字（半角），不能用全角。

setup.py

```
from setuptools import find_packages, setup

package_name = 'lanedetection'

setup(
    name=package_name,
    version='0.0.0',
    packages=find_packages(exclude=['test']),
    data_files=[
        ('share/ament_index/resource_index/packages',
            ['resource/' + package_name]),
        ('share/' + package_name, ['package.xml']),
    ],
    install_requires=['setuptools'],
    zip_safe=True,
    maintainer='root',
    maintainer_email='root@todo.todo',
    description='TODO: Package description',
    license='TODO: License declaration',
    tests_require=['pytest'],
    entry_points={
        'console_scripts': [
        'lanedetection =lanedetection.lanedetection:main',
        ],
    },
)

```

3）对功能包进行编译

cd /userdata/dev_ws

colcon build --packages-select lanedetection --symlink-install

--packages-select，只编译该功能包，

--symlink-install，install 和 src 同步，修改源程序后，不需要再编译了。

```
root@ubuntu:/userdata/dev_ws/src/originbot# cd /userdata/dev_ws
root@ubuntu:/userdata/dev_ws# colcon build --packages-select lanedetection --symlink-install
Starting >>> lanedetection
Finished <<< lanedetection [7.52s]

Summary: 1 package finished [10.5s]
root@ubuntu:/userdata/dev_ws#
```

4）修改 lanedetection.py 源程序

```
def pct_callback(self, msg):
    self.pct=msg
    #self.pct.linear.x  推理出来的小车车速
    #self.pct.angular.z 推理出来的小车转角

    #self.twist.linear.x =      控制小车速度
    #self.twist.angular.z =     控制小车转角


    self.cmd_vel_pub.publish(self.twist)
```

5）开 3 个窗口，运行小车车道巡线

①ros2 launch originbot_bringup originbot.launch.py use_camera:=true
②ros2 run line_follower_perception line_follower_perception
③ros2 run lanedetection lanedetection